hcs-workcell

Setting up the cell and running it

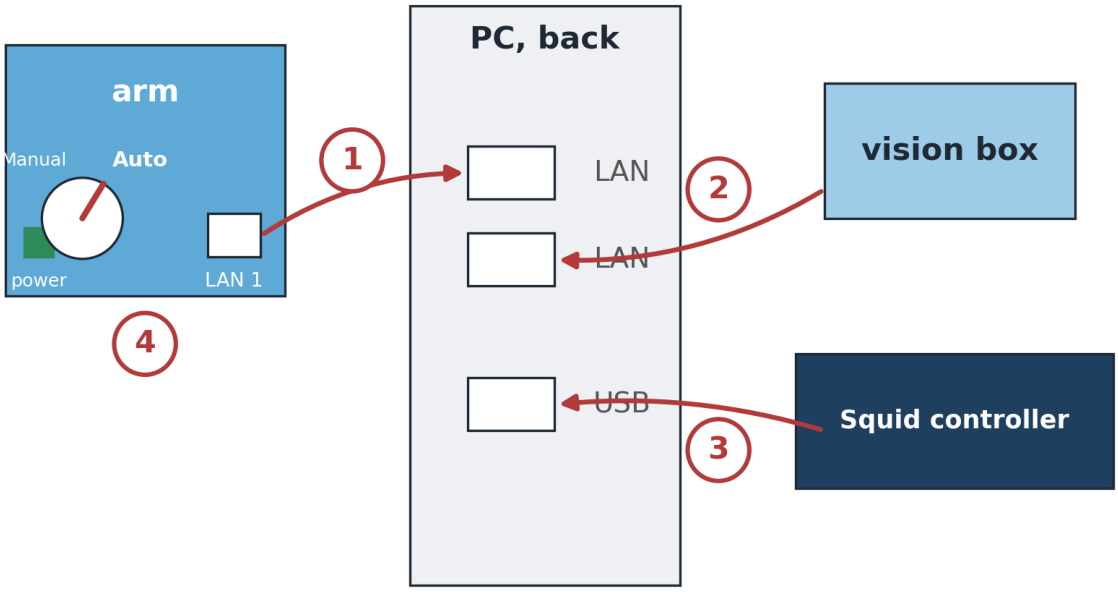
Two programs. setup: Setup, Cameras, Teach, Test. run: Start, STOP.

0. Cables
1. Setup
2. Cameras
3. Teach
4. Test
5. Run
6. If something stops

STOP, on screen, at any time: the arm holds still.

0 Cables

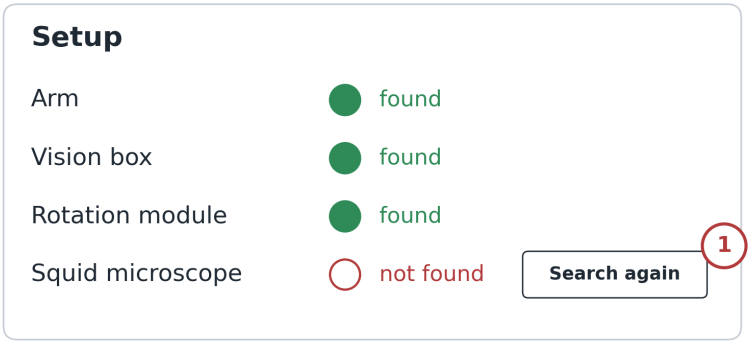
Plug in the three cables.
Switch the arm on. Key at Auto.



1 arm, port LAN 1, to the PC
2 vision box to the PC
3 Squid controller USB to the PC
4 arm power on, key at Auto

1 Setup

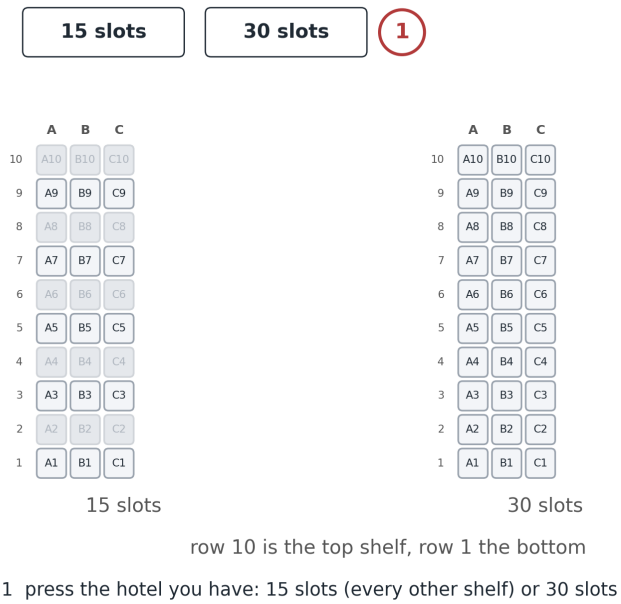
Open the setup program. Setup, Cameras, Teach and Test live in it; run is the other program.
It finds the arm and the vision box by itself. Nothing to type. Every row green: go on.



1 a row not green: power, key at Auto, that device's cable, then Search again

1.1 The hotel

Press the hotel you have: 15 slots or 30 slots.

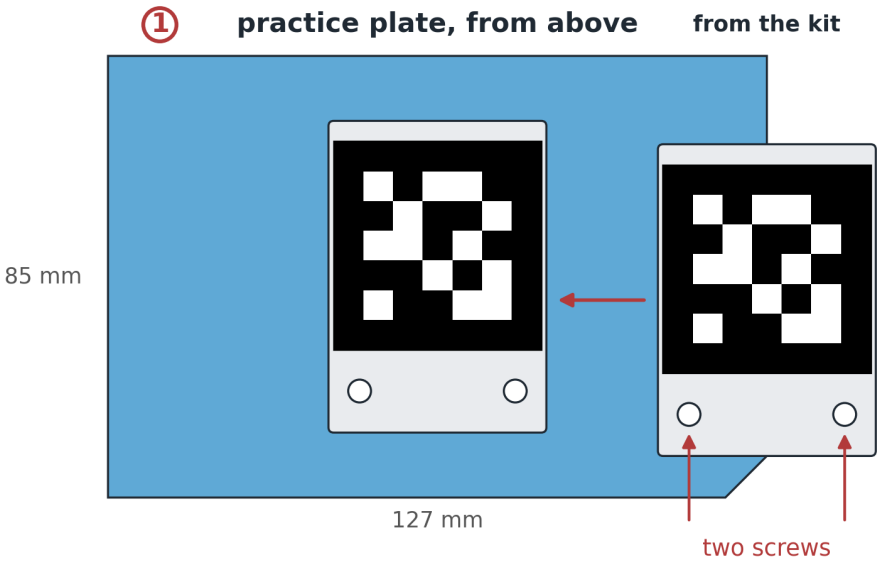


row 10 is the top shelf, row 1 the bottom

1 press the hotel you have: 15 slots (every other shelf) or 30 slots

2 Cameras

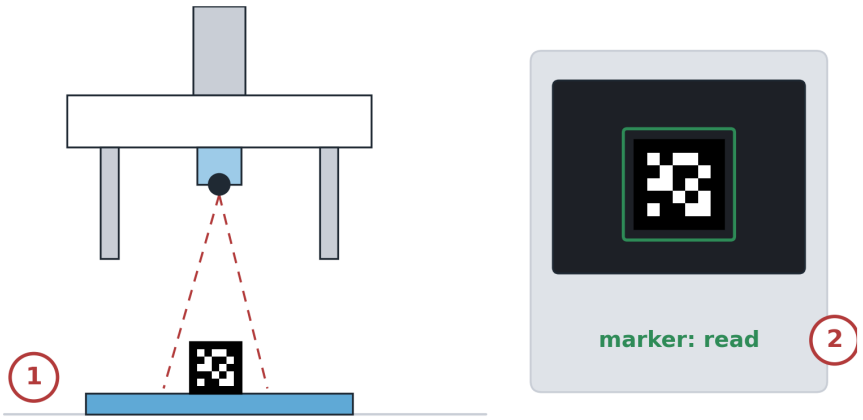
Screw the marker tab from the kit onto the practice plate.



1 screw the tab onto the practice plate. The box reads the marker

2.1 Does it read the marker?

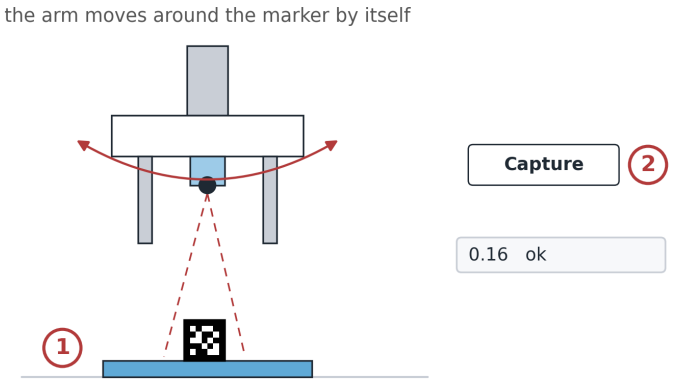
Put the practice plate with the marker under the camera. The screen says it reads the marker.
If not: clean the lens, or move the plate closer.



1 the practice plate with the marker under the down camera
2 the screen says it reads the marker
3 not read: clean the lens, or move the plate closer

2.2 Capture

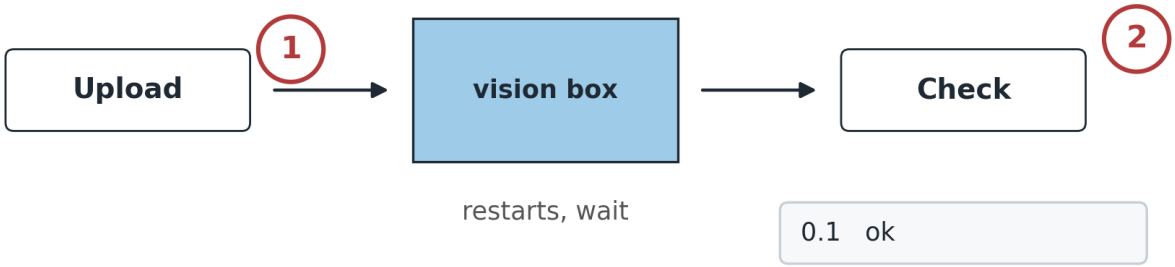
Put the practice plate with the marker on the bench. Move the arm so the marker is in the frame. Press Capture.
The arm moves around the marker by itself and takes its pictures. Then a number and ok, or redo.



1 the practice plate with the marker, flat on the bench, the marker in the frame
2 press Capture. The arm moves by itself; then a number and ok, or redo

2.3 Upload, Check

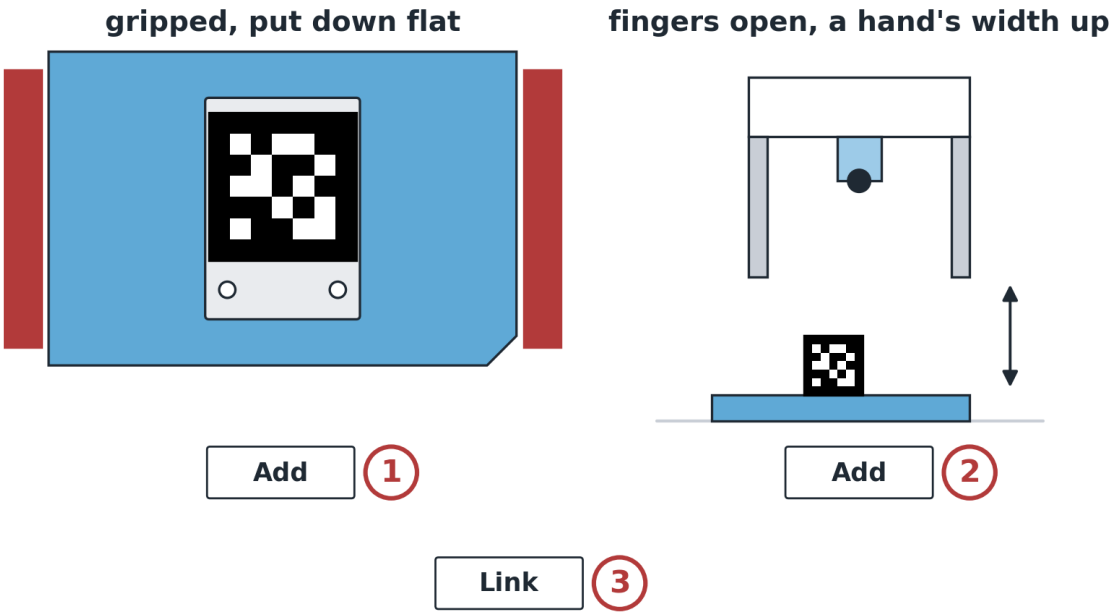
Press Upload: the box restarts. Then press Check: a number and ok.



1 press Upload, wait for the box to come back
2 press Check: a number and ok

2.4 Add, Add, Link

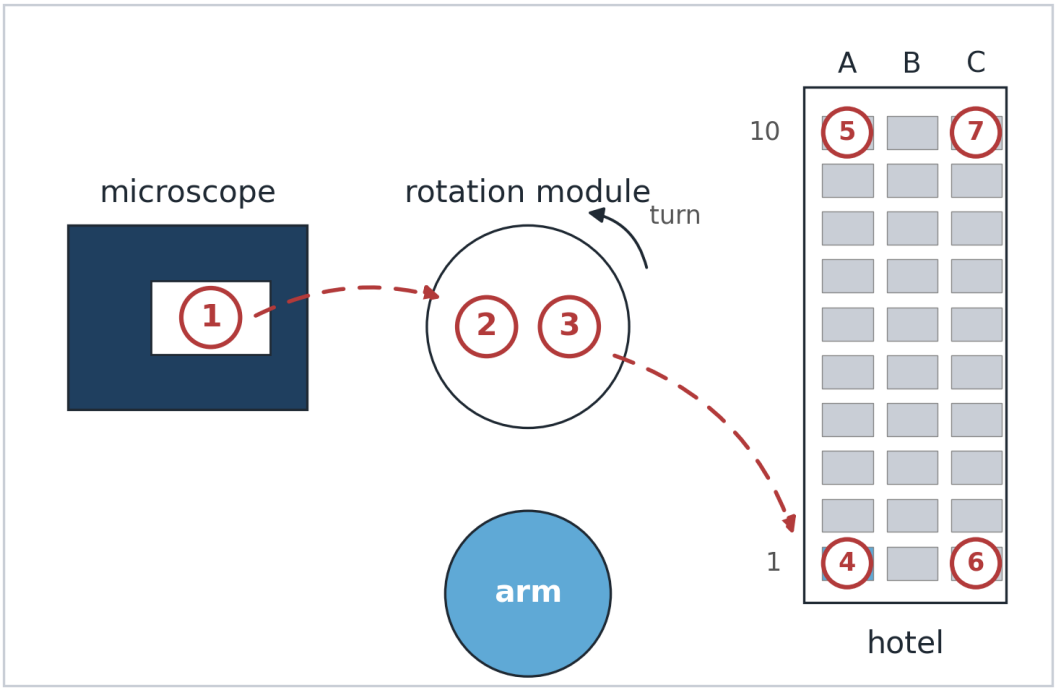
Grip the practice plate with the marker, put it down flat, press Add. Open the fingers, hover a hand's width above it, press Add. Press Link.
Once with the short edges in the fingers, once with the long edges.



3 Teach

One list. Put the practice plate where the row says, hover the hand over it so the marker is in view, press Record.
Go down the list. The way you drag the arm between two Records is the route it will use.

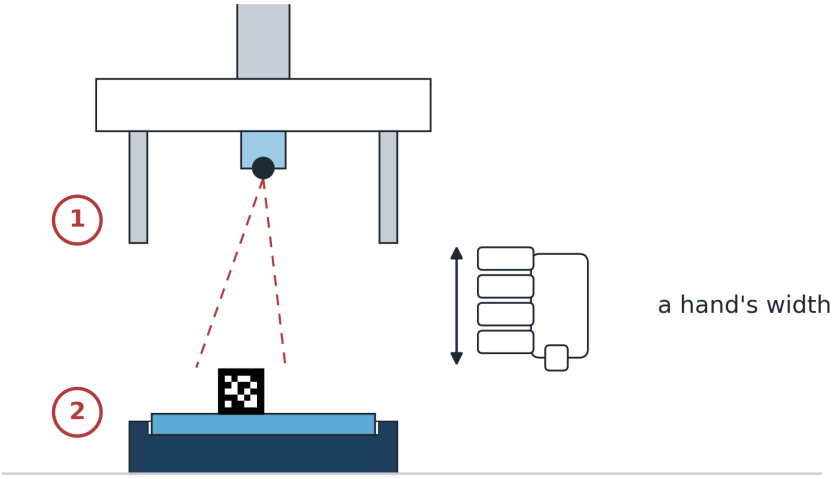
put the practice plate at each mark, hover the hand over it, press Record



1 the microscope holder, plate in the holder
2 rotation module at 0, plate seated, short edges in the fingers
3 rotation module at 90, plate seated, long edges in the fingers
4 hotel A1 5 hotel A10 6 hotel C1 7 hotel C10
dashed: the drag between two Records is the route the arm will use

3.1 Hover

A hand's width above the plate. The marker whole in the down camera's picture.

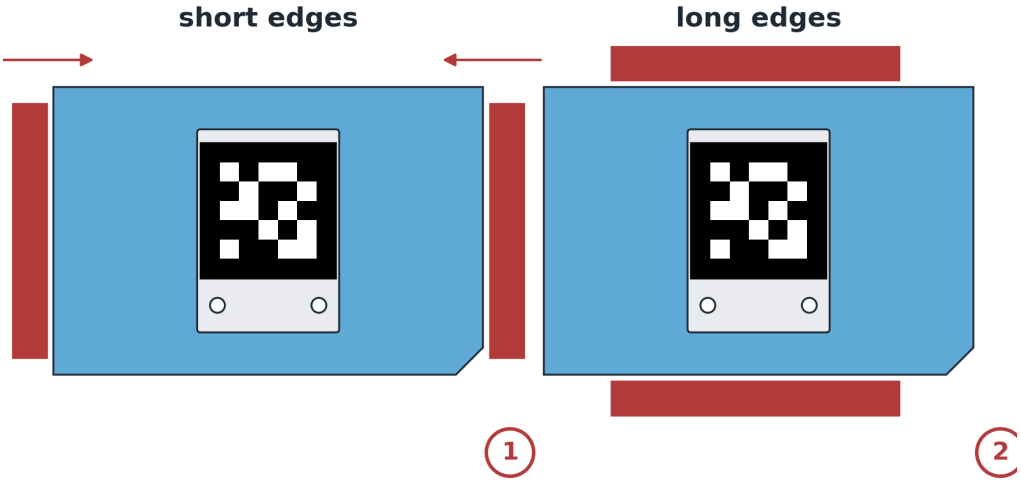


1 a hand's width up, the marker whole in the picture: press Record

2 the plate rests where it will rest in a run

3.2 Grip

Plate between the fingers. Press Measure with the short edges in the fingers, then again with the long edges.

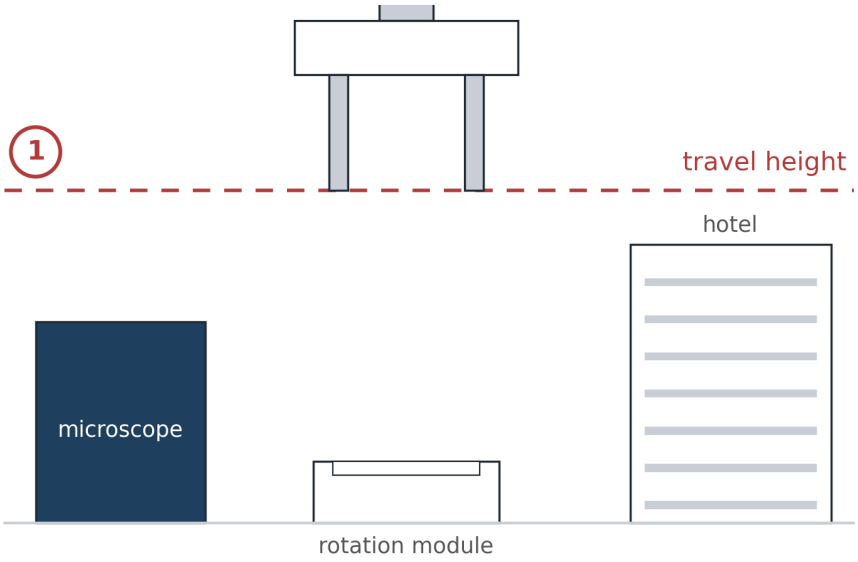


1 short edges, the two ends: microscope holder, rotation module at 0

2 long edges, the two sides: rotation module at 90, hotel

3.3 Travel height

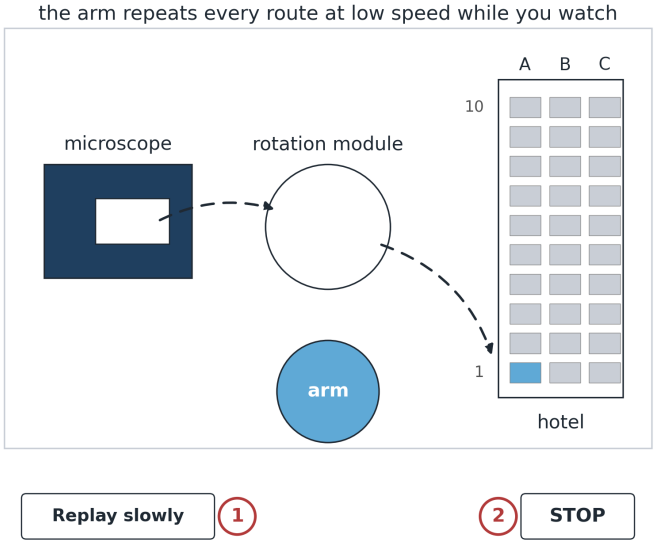
Lift the hand until it clears everything on the bench. Press Record.



1 lift the hand until it clears everything on the bench, press Record

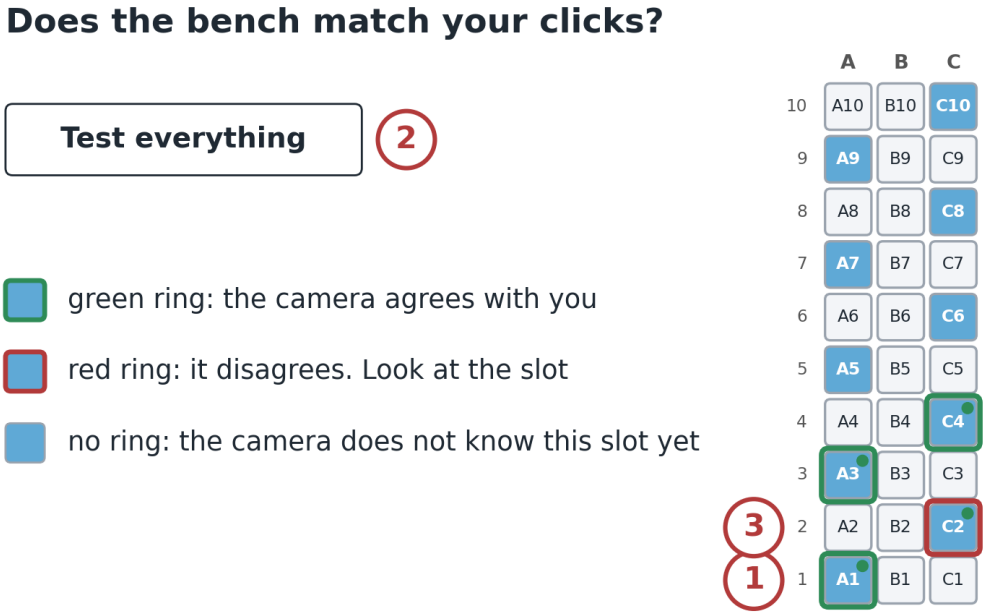
3.4 Replay slowly

Press Replay slowly. The arm repeats every route at low speed while you watch.
Done when no row in the list is red.



4 Test

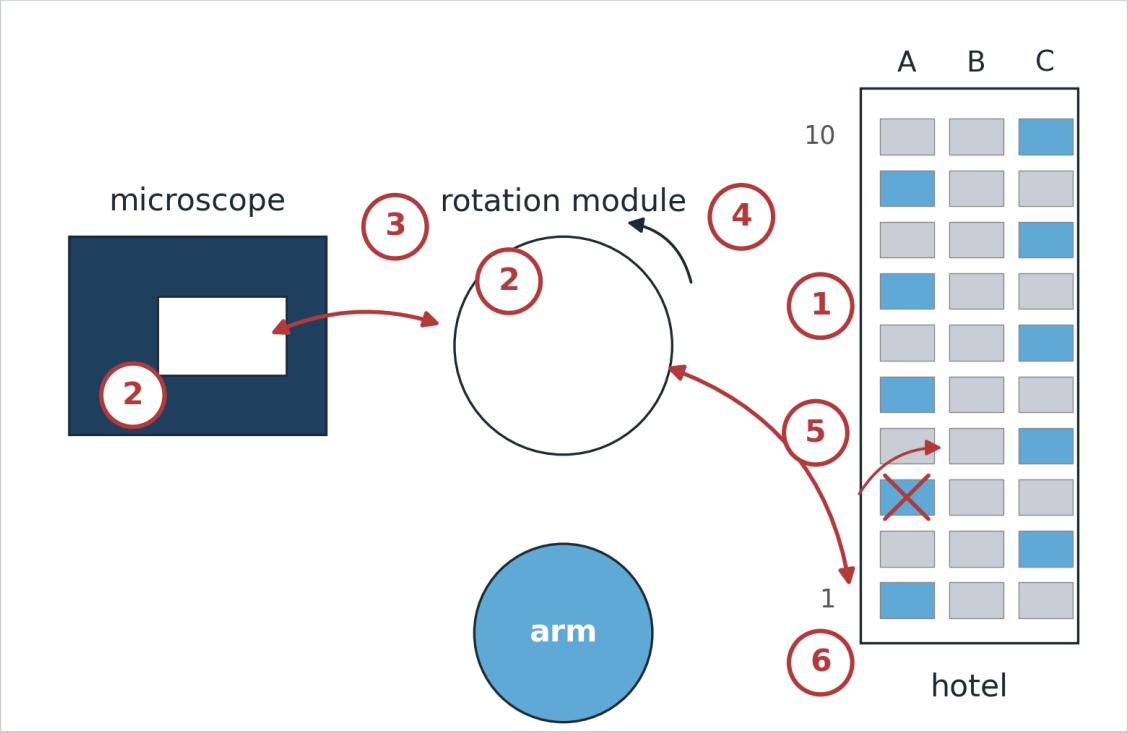
Click the slots that really hold a plate. Press Test everything.
Watch. It goes round and round until you press STOP. A red ring on a slot: go and look at it.



1 click the slots that really hold a plate
2 one button. The arm does the whole test on its own, slowly
3 a red ring: go and look at that slot

4.1 What the test does

The arm does all of it on its own, slowly, in this order.
Done when every plate is back in its own slot and the screen says finished.



1 looks into every slot and compares with your clicks

2 reads the marker at every taught position

3 drives the path slowly between the microscope and the rotation module

4 turns the rotation module, camera check first

5 orders a plate into a taken slot on purpose, backs out, uses another slot

6 takes every plate to the microscope and back, in turn: A1, C2, A3, ...

5 Run

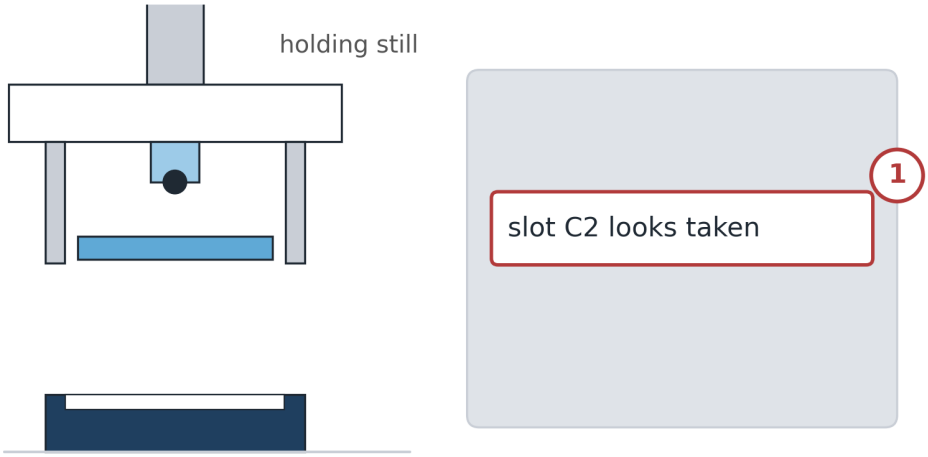
When the tests pass, open the run program and press Start.



1 the arm looks into every slot, shows what it found, then runs every plate in turn
2 STOP at any time. The arm holds still. Start again continues

6 If something stops

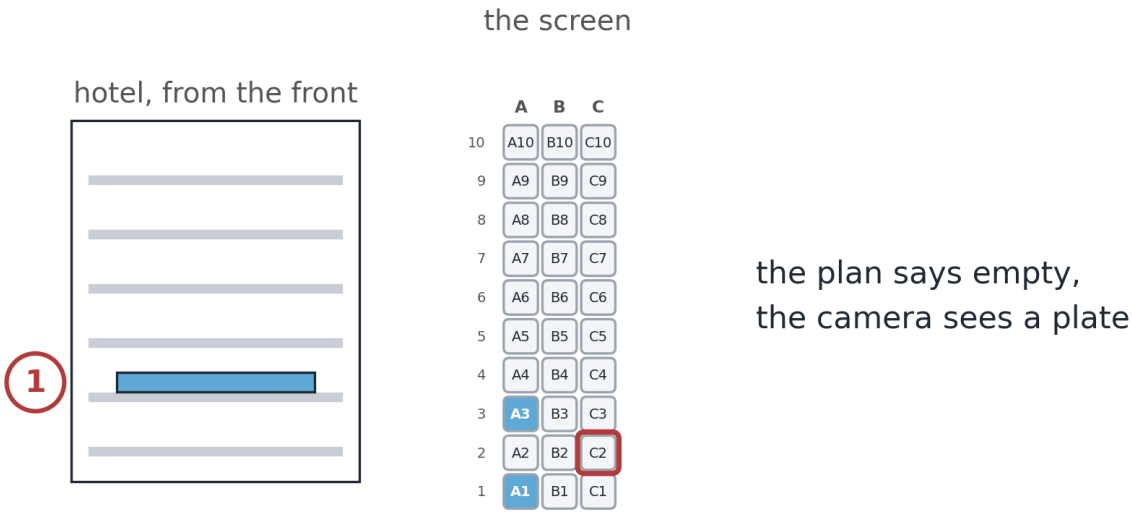
The arm holds still. Read the one line on screen. It is one of the three below.



1 one line on screen says what the arm saw. Read it, then the next three pages

6.1 A slot that looks taken

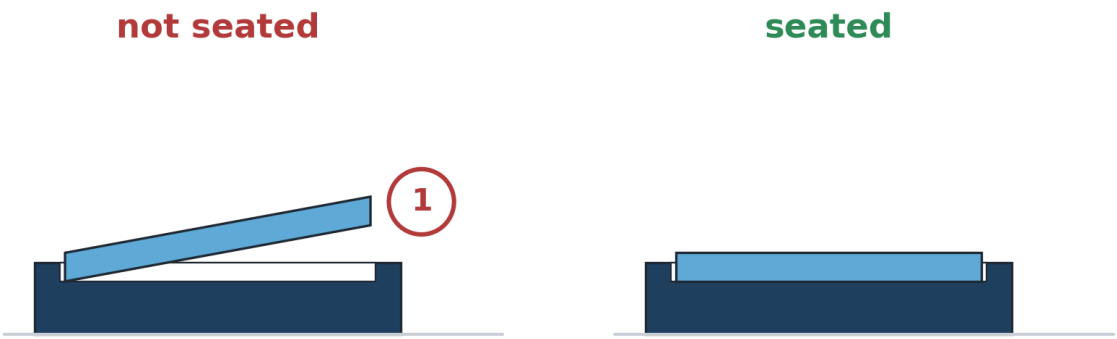
Take the plate out of that slot, or click the slot in Test. Then Start.



1 a plate someone left in a slot: take it out, or click that slot in Test. Then Start

6.2 A plate that does not look seated

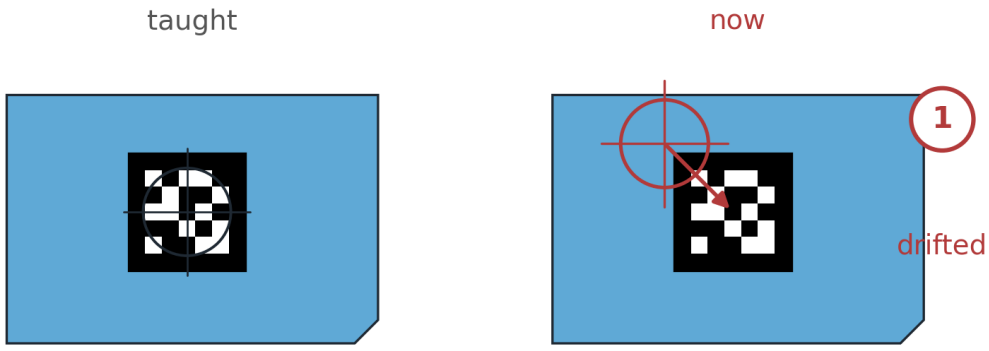
Seat the plate by hand, flat in the recess. Then Start.



1 on a corner, on the rim, or turned: seat it flat by hand. Then Start

6.3 A position that drifted

Something was bumped. Teach: find that row, Record again. Then Start.



1 something was bumped: Teach, that row, Record again. Then Start